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An efficient remote driving shift control method of unmanned heavy tracked vehicles based on manned data mining

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After the unmanned transformation of the transmission system of a heavy tracked vehicle, there are problems of low shift smoothness and long shift time during remote driving. Aiming to realize the smooth shift of tracked vehicles by remote driving, this study mines the manned driving data of the vehicle and develops an auxiliary decision-making model for the shift timing of remote driving. In addition, an integrated shift control strategy of tracked vehicle power transmission based on a hierarchical finite state machine is designed. The effectiveness of the proposed remote control shift strategy for tracked vehicles is verified by bench test and real vehicle test. During the entire shifting process, the average shift impact of the tracked vehicle is 9.02 $\mathrm{m/s^3}$, the average shift time is 4.6 s, and the vehicle has good ride comfort。 The research results can provide useful guidance and lay a foundation for the research of driving control and trajectory tracking control of this type of heavy tracked vehicle after unmanned transformation.

Keywords Unmanned heavy tracked vehicles, Shift smoothness control, Data mining

Developing heavy unmanned tracked vehicles using unmanned transformation of a transmission system of the existing heavy-duty tracked vehicles represents an effective method for equipping vehicles with remote control and unmanned driving functions. The heavy tracked vehicles studied in this paper include traditional transmission devices, such as diesel engines, fixed-axis gearboxes, dry clutches, etc. During the unmanned transformation, the original mechanical control mechanism was replaced by a hydraulic control mechanism. When a vehicle is driving remotely, its controller controls a clutch, gear, and throttle control mechanism to complete the shift automatically according to the given remote-control shift command. After the unmanned transformation of the vehicle, the vehicle has the basis of remote driving and unmanned driving. Therefore, future research should put forward an integrated control strategy of power and transmission and realize a smooth shift of vehicle remote control[1].

Recently, advanced intelligent algorithms such as deep learning[2] and reinforcement learning[3] have been widely applied in the field of unmanned vehicle driving. Yang[4] addressed the overfitting problem in current path planning algorithms by proposing a path planning algorithm based on improved deep reinforcement learning. They also verified the effectiveness of this method through experiments. Wang[5] developed a deep Q-network (DQN) model to solve the route decision-making problem of unmanned vehicles in complex traffic environments; this model takes dynamic traffic information as input and typical driving decisions as action outputs. Simulation results show that this model has significant potential for enabling unmanned vehicles to learn challenging driving decisions. Wu[6] focused on the driving safety of unmanned vehicles in highly complex environments and proposed a risk-aware reinforcement learning method based on driving task decomposition. The method aims to minimize risks from various sources. Chen[6] integrated deep learning into the perception and localization of autonomous platforms and proposed a deep learning-enhanced visual SLAM (Simultaneous Localization and Mapping) framework. This framework has improved navigation accuracy by 5%. Han[7] addressed the limitations of multi-sensor data assimilation in the environmental perception process of unmanned ground vehicles (UGVs) and proposed a multi-sensor fusion method based on long short-term memory (LSTM) networks. This method alleviates the attenuation of environmental perception information during the navigation of unmanned vehicles. In general, the application of intelligent algorithms in the field of unmanned driving is mostly concentrated on

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environmental perception and behavioral decision-making. Research on vehicle drive control remains relatively scarce, especially for heavy tracked vehicles equipped with diesel engines.

Research on vehicle drive systems mostly focuses on the integrated power transmission control of vehicles[8–10]. Zhang[11] studied the shift control of pure electric vehicles and proposed a motor coupling control strategy and a shift coordination control method for the vehicle shift process, which could effectively improve the shift quality of pure electric vehicles. Mishra[12] designed an integrated power transmission shift controller based on a gain scheduling observer using linear variable parameter control technology. The experimental results showed that the scheme could effectively control the shift inertia phase under different throttle opening conditions. Oncken[13] proposed a predictive control strategy for optimal mode selection and a power transmission integrated shift control method for multi-mode hybrid electric vehicles. The experimental results demonstrated that the proposed method could reduce the energy consumption of electric vehicles by 4%–10% under standard conditions. For mild hybrid vehicles, Kargar[14] established an enhanced model that includes vehicle external dynamics and powertrain dynamics, and proposed a customized control strategy based on approximate dynamic programming, which effectively improved the vehicle's shift fuel economy. To address the poor shift quality of heavy-duty vehicles, Wang[15] proposed a power downshift control method based on adaptive sliding mode control, which effectively improved the shift quality of heavy-duty vehicles. Considering the same type of tracked heavy vehicle as that studied in this work, Liu[16] proposed a prediction model of gearing time based on the CART decision tree algorithm, which could improve the reliability of integrated control of vehicle power transmission. Wang[17] developed a state analysis method of a planetary transmission mechanism for the vehicle shift process and designed an integrated shift control strategy for power transmission. Zhang[18] applied the time slice rotation scheduling method to design the automated mechanical transmission(AMT)shift program architecture on a single-chip microcomputer. Wang[19] designed an electronic control system for an engine and proposed an incremental PID control strategy, which could realize the double closed-loop control of the fuel supply gear rod's position and engine speed. Although the solutions proposed in the above-mentioned studies can effectively improve the shift success rate of the tracked vehicle, these studies have not performed research on the shift smoothness and reliability of the tracked vehicle under remote driving conditions.

Motivated by the previous research and aiming at the problem of low shift smoothness after the modification of the transmission system of heavy tracked vehicles, this work studies the integrated control of power and transmission of a vehicle during remote driving. First, the manned driving data of a vehicle are collected, and an auxiliary decision-making model of remote driving shift timing is established based on the support vector machine (SVM) algorithm to ensure that the tracked vehicle performs shift operation at the right time and improve the shift smoothness of the vehicle. Then, an integrated shift control strategy of a tracked vehicle's power transmission based on the hierarchical finite state machine (HFSM) is designed to improve the reliability and safety of vehicle remote control shift. Finally, the proposed shift control strategy is verified by tests. The proposed method can not only provide a reference for the driving control of heavy-duty unmanned vehicles equipped with diesel engines after unmanned modification, but also lay a foundation for further realizing automatic shifting of heavy-duty tracked vehicles and longitudinal speed tracking control for autonomous driving.

The remainder of this thesis is organized as follows. In Sect. Shift auxiliary Decision-making model, a vehicle shift auxiliary decision-making model is established, which consists of two parts: data mining of shift timing based on the K-Means algorithm, and the development of the vehicle shift auxiliary decision-making model based on the SVM algorithm. In Sect. Integrated shift control strategy for tracked vehicle power and transmission systems based on HFSM, an integrated power and transmission control strategy for vehicles based on the HFSM is proposed. In Sect. Experimental verification, the proposed shift strategy is verified through bench tests and real-vehicle tests. Specifically, the bench tests validate the control effects under various scenarios, including the vehicle's second-gear start, shift-up, return to the original gear after a failed shift-up, and fault handling; the real-vehicle tests verify the control effect of the vehicle's continuous shifting. Finally, in Sect. Conclusion, conclusions regarding the controller are presented.

Shift auxiliary Decision-making model

In the process of driving a tracked vehicle by remote control, it is challenging for a driver to perceive the vehicle's state in real time because the driver is not in the cockpit, and the choice of shift timing is prone to deviation. The shift timing refers to the time when a driver starts the clutch separation operation, and its selection has a significant impact on improving the success rate of the shift, as well as the shift quality. To ensure the shift success rate of a tracked vehicle's remote driving and improve the shift smoothness, this paper has proposed an auxiliary decision-making method of shift timing for remote driving based on data mining of manned driving training.

Data mining of shift timing decision based on K-Means algorithm

To mine the manned driving data, it is necessary to collect the manual driving data of the vehicle first. Sensors are installed at the engine and throttle of a tracked vehicle to collect the manned driving data, and an engine speed n_e and a throttle opening β_t at the time of shift operation are extracted from the data. Based on the K-Means algorithm[20], a driver's shift timing is analyzed, and the number of cluster datasets N_k is set to 3. The K-Means algorithm can rapidly cluster the driving operation information on different drivers to form an obviously separated cluster dataset. The objective function of the K-Means algorithm is:

$$J_K = \sum_{i_K=1}^{N_K} \sum_{k_K=1}^{M_K} r_{i_K k_K} \parallel x_{i_K} - u_{k_K} \parallel^2 \tag{1}$$

where k_K is the cluster dataset, and u_{k_K} is the mean vector of k_K.

In Eq. (1), when a sample x_{i_K} is divided into the cluster dataset k_K, $r_{i_K k_K} = 1$; otherwise, $r_{i_K k_K} = 0$, which can be expressed as follows:

$$r_{i_K k_K} = \begin{cases} 1 & , x_{i_K} \in k_K \\ 0 & , x_{i_K} \notin k_K \end{cases} \tag{2}$$

By defining an objective function J_k at a mean vector u_{k_K}, the mean vector expression corresponding to the optimal objective function can be derived as follows :

$$\frac{\partial J_K}{\partial u_{i_K}} = \sum\nolimits_{i_K=1}^{N_K} \sum\nolimits_{k_K=1}^{M_K} r_{i_K k_K} \left(x_{i_K} - u_{k_K}\right) = \sum\nolimits_{i_K=1}^{N_K} r_{i_K k_K} \left(x_{i_K} - u_{k_K}\right) = 0 \tag{3}$$

The expression of mean vector u_{k_K} corresponding to the optimal objective function can be expressed as follows:

$$u_{i_K} = \frac{\sum\nolimits_{i_K=1}^{N_K} r_{i_K k_K} x_{i_K}}{\sum\nolimits_{i_K=1}^{N_K} r_{i_K k_K}} \tag{4}$$

Figure 1 shows the relationship between the n_e and β_t parameters at the shift time when a driver performs the upshift operation. Figure 1(a) shows the throttle opening and engine speed at the starting and shifting times of the second gear of the vehicle. As displayed in Fig. 1(a), most drivers perform a slight throttle action when starting in the second gear. At this time, the engine speed is in the range from 650 $\mathrm{r/min}$ to 850 $\mathrm{r/min}$, and the engine speed of the cluster center is n_e^c=721 $\mathrm{r/min}$. Figure 1(b) shows the variation of the values of n_e and β_t during the vehicle's 2nd-to-3rd gear shift. Before shifting gears, the driver will take actions such as increasing the throttle opening and raising the vehicle speed. At this time, the β_t value is generally from 30% to 60%, the n_e value is in the range of 1,200 $\mathrm{r/min}$–1,500 $\mathrm{r/min}$, and the engine speed of the cluster center is n_e^c=1431$\mathrm{r/min}$. Figure 1(c) shows the variation of the values of n_e and β_t during the vehicle's 3rd-to-4th gear shift. At this time, the β_t value is generally approximately 60%, the n_e value is from 1,400 $\mathrm{r/min}$ to 1,700 $\mathrm{r/min}$, and the

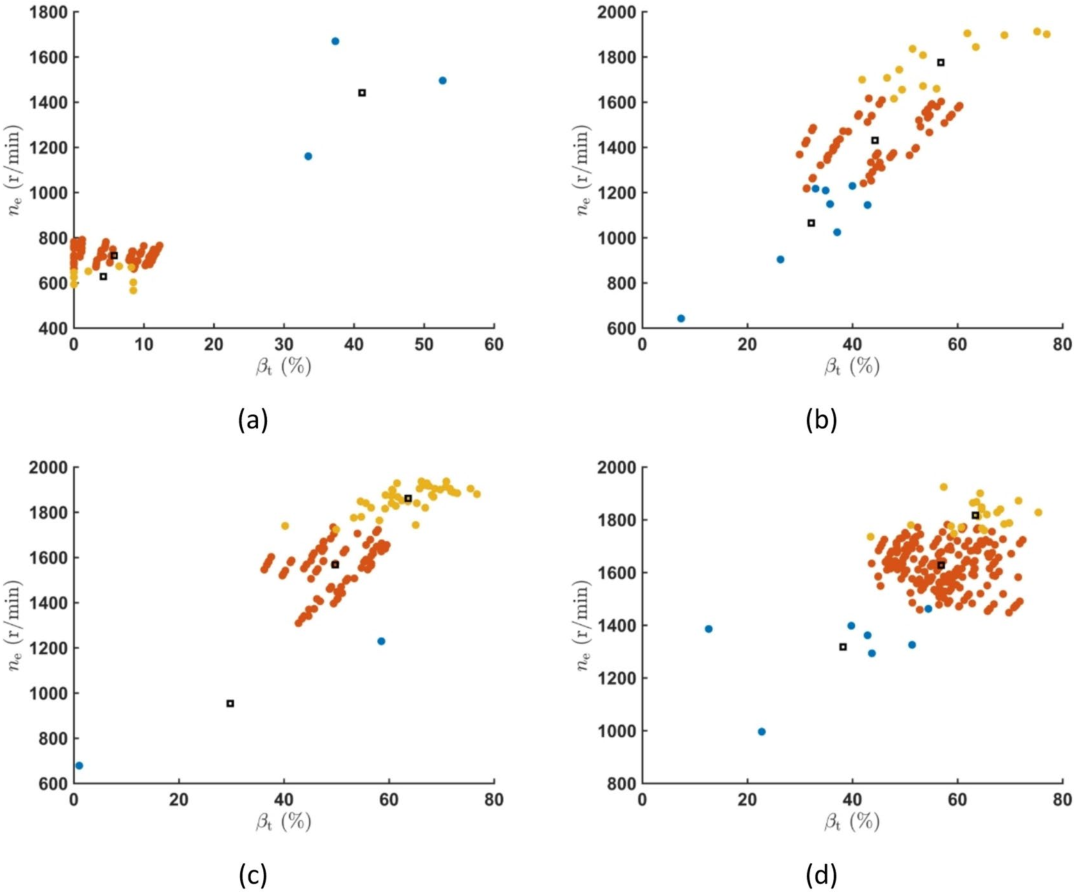

Fig. 1. The relationship between the n_e and β_t parameters at the shift time when a driver performs the upshift operation: (**a**) the driver starts the vehicle in the second gear; (**b**) the driver performs the two- to three-gear operation; (**c**) the driver performs the three- to four-gear operation; (**d**) the driver performs the four- to five-gear operation.

cluster center engine speed is n_e^c=1,567 r/min. Figure 1(d) shows the variation of the values of n_e and β_t during the vehicle's 4th-to-5th gear shift. At this time,, the throttle opening generally has a value from 60% to 80%, the n_e value is in the range of 1,500 r/min–1,800 r/min, and the engine speed of the cluster center is n_e^c=1,628 r/min. In this phase, the driver will perform the throttle opening operation when the gear is up. With the increase in the gear value, the engine speed will gradually increase at the time of shift. It should be noted that some drivers will increase the engine speed to a larger value during the upshift operation. A larger engine speed is likely to increase the shift impact and reduce the shift quality.

Figure 2 presents the $n_e$and β_t values at the time the driver performs the downshift operation. Unlike the upshift operation, in the downshift operation, a driver mostly reduces the throttle opening to reduce the engine speed. The reason for this action is that a vehicle's downshift transmission ratio increases, and the speed of the passive part of the clutch decreases after the gear is completed. To reduce the friction work and effect during the clutch engagement, the engine speed should not be too high. When the driver performs a 5th-to-4th gear shift, the throttle opening is generally reduced to decrease the engine speed to the range of 1,100 r/min–1,400 r/min. Similarly, the engine speed is mostly reduced to the range of 1,000 r/min–1,300 r/min when the 4th-to-3rd and 3rd-to-2nd operations are performed. During the two-gear shift operation, the engine speed is generally at the lowest stable speed level.

Auxiliary Decision-making model of shift timing based on SVM algorithm

In this study, the manual driving data of tracked vehicles are used, the driving data shift process fragments are screened, and the shift timing assistance model is designed. The shift process fragment information mainly includes the current gear value $gear_{\mathrm{cur}}$, the target gear value $gear_{\mathrm{cmd}}$, an engine speed n_e, a throttle opening β_t, and a vehicle speed v. By setting the evaluation criteria, the rationality of the shift timing decision is judged. The shift timing decision rationality flag, denoted by $flag_g^d$, is set to mark the shift segment. The flowchart of the specific process is displayed in Fig. 3.

The selection of shift timing has an important influence on shift quality. The shift quality is used as a judgment criterion of the shift timing decision rationality, and the collected driving data are classified. The judgment criteria mainly include a shift duration t_g, an engine speed fluctuation during shift Δn_{eg}, the maximum jerk during shift j_g, and an average shift speed $\overline{v}_g$. The maximum impact degree takes the standard value of the

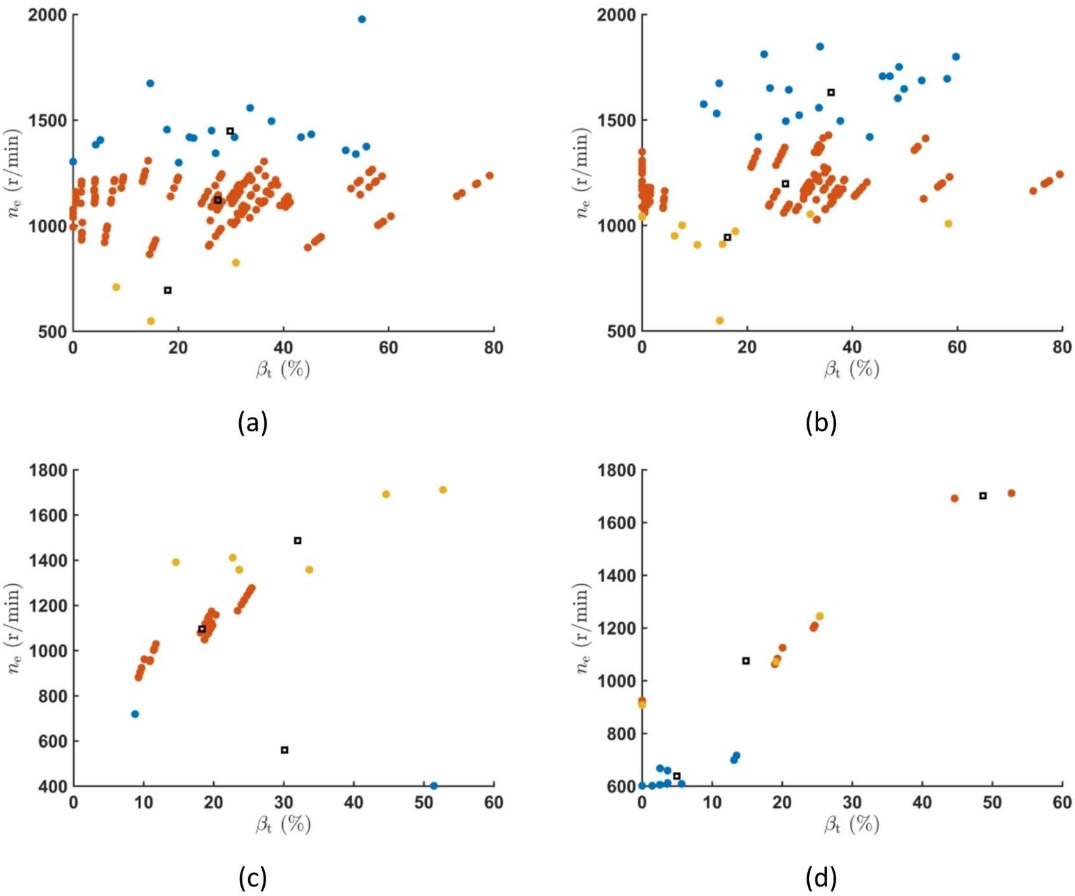

Fig. 2. The n_e and β_t values at the time of shift when a driver performs the downshift operation: (**a**) the driver performs the five- to four-gear operation; (**b**) the driver performs the four- to three-gear operation; (**c**) the driver performs the three- to two-gear operation; (**d**) the driver performs the two-gear shift operation.

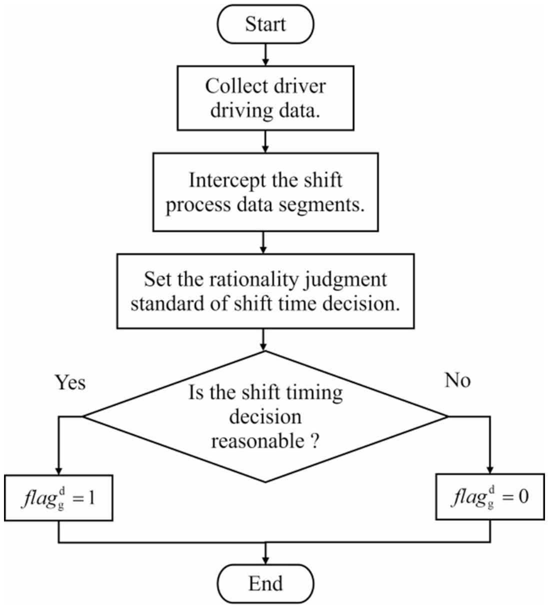

Fig. 3. The flowchart of the marking process of shift time decision rationality.

g_{cur}	g_{cmd}	$\widehat{t}_g$ (s)	$\Delta\,\widehat{n}_{eg}$ (r/min)	$\widehat{\overline{v}}_g$ (km/h)
0	2	5.6	209	1.29
2	3	4.3	562	10.1
3	4	4.1	811	18.23
4	5	3.7	723	22.96
5	4	4.35	647	25
4	3	4.4	578	18.67
3	2	5.23	538	12.08
2	0	3.25	132	5.42

Table 1. The shift timing judgment standard values.

impact degree of heavy tracked vehicles as $\widehat{j}_g = 31.6\mathrm{m/s^3}$. The standard values of the shift duration $\widehat{t}_g$, engine speed fluctuation $\Delta\,\widehat{n}_{eg}$, and shift average speed $\widehat{\overline{v}}_g$ are set as the average value of the relevant data analyzed in Sect. Data mining of shift timing decision based on K-Means algorithm; the specific values are presented in Table 1.

The rationality marking method of the shift time decision of a shift segment can be defined as follows:

$$flag_g^d = \begin{cases} 1 & ,t_g \le \widehat{t}_g, j_g \le \widehat{j}_g, \Delta\,n_{eg} \le \Delta\,\widehat{n}_{eg}, \overline{v}_g \ge \widehat{\overline{v}}_g \\ 0 & ,others \end{cases} \tag{5}$$

After marking the rationality of the shift decision-making result for each shift-process segment, the SVM algorithm is employed to design an auxiliary decision-making model of shift timing. The SVM algorithm is a supervised learning algorithm which has been widely used to solve binary classification problems. Its basic principle is to find the maximum interval hyperplane between the sample points of various datasets. The specific linear equation is as follows:

$$w_S^T x_S + b_S = 0 \tag{6}$$

where w_S is the normal vector of the hyperplane, and b_S is the distance from the hyperplane to the origin of the coordinate axis.

The distance from any point x_{S_i} to $w_S^T x_{S_i} + b_S = 0$ in the hyperspace, denoted by r_S, can be calculated by:

$$r_S = \frac{\left|w_{S_i}^T x + b_S\right|}{\parallel w_S \parallel} \tag{7}$$

Next, define a point $y_{S_i} = 1$ in the distance hyperplane of the sample point of the dataset, and $y_{S_i} = -1$ when the distance is negative. Then, the distance function distance expression of the sample point of the data set from the hyperplane is:

$$y_{S_i}(w_S^T x_S + b_S) \tag{8}$$

By combining Eqs. (7) and (8), the value of distance r_S can be obtained by:

$$r_S = \frac{y_{S_i}\left(w_{S_i}^T x + b_S\right)}{\parallel w_S \parallel} \tag{9}$$

The SVM hyperplane segmentation diagram is shown in Fig. 4. The support vector is located on the support plane, and the function distance is 1. The other sample points are located outside the support plane, and the distance from the hyperplane function is greater than 1. Setting the distance of the sample point function as a constraint condition, the optimal hyperplane can be expressed as follows:

$$\underset{w_S, b_S}{min} \frac{1}{2} \parallel w_S \parallel^2 s.t.\ y_{S_i}(w_S^T x_S + b_S) \leq 1 \tag{10}$$

Further, the Lagrangian multiplier method is used to transform the constraint condition, and the Lagrangian function about the hyperplane is derived. The specific expression is as follows:

$$L(w_S, b_S, \alpha_S) = \frac{1}{2} \parallel w_S \parallel^2 + \sum_{i=1}^{m_S} \alpha_{S_i} (1 - y_{S_i}(w_S^T x_{S_i} + b_S)) \tag{11}$$

where m_S is the number of sample points, and α_S is the Lagrange multiplier.

The partial derivatives of w_S and b_S are obtained by:

$$\begin{cases} \frac{\partial L}{\partial w_S} = w_S - \sum_{i=1}^{m_S} \alpha_{S_i} x_{S_i} y_{S_i} = 0 \\ \frac{\partial L}{\partial b_S} = \sum_{i=1}^{m_S} \alpha_{S_i} y_{S_i} = 0 \end{cases} \tag{12}$$

By transforming Eq. (11) into strong duality and substituting Eq. (11) into Eq. (12), the following expression can be obtained:

$$L(w_S, b_S, \alpha_S) = \underset{\alpha_S}{max} \sum_{i=1}^{m_S} \alpha_i - \frac{1}{2} \sum_{i=1}^{m_S} \sum_{j=1}^{m_S} \alpha_{S_i} \alpha_{S_j} y_{S_i} y_{S_j} x_{S_i}^T x_{S_j} \tag{13}$$

The Sequential Minimal Optimization (SMO) algorithm is used to solve Eq. (13) and calculate the optimal Lagrange multipliers. The obtained optimal Lagrange multipliers are then substituted into Eq. (11) to solve for the optimal hyperplane.

By collecting the manned driving data on the tracked vehicle, a total of 2,160 shift process segments are screened. Then, each shift segment is marked by the rationality standard of shift timing decision. The values of the engine speed n_e, throttle opening β_t, current gear $gear_{\mathrm{cur}}$, and target gear $gear_{\mathrm{cmd}}$ in the shift process fragment are selected as decision input characteristic parameters of the shift time. The shift process flag $flag_g^d$ is used as a rationality output feature of the shift time decision. The SVM algorithm is employed to train the driving data, and the auxiliary decision model of shift timing is constructed. Further, 85% of the collected shift process segments is used for model training, and the remaining 15% is used for model validity testing. Among them,

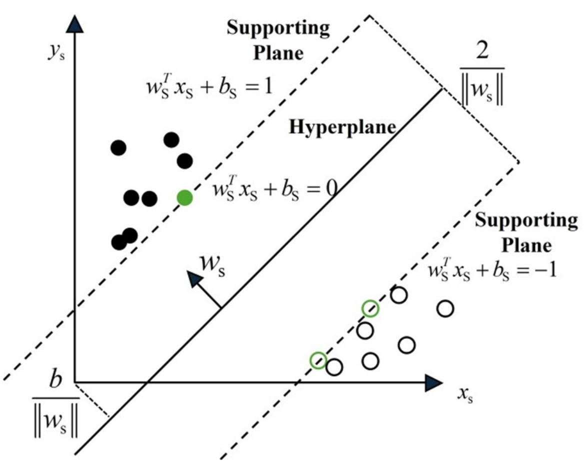

Fig. 4. The SVM hyperplane segmentation diagram.

Fig. 5. The training results of the shift timing assistant decision model based on the SVM algorithm: (**a**) The prediction results for the training data (**b**) The prediction results for the test data set.

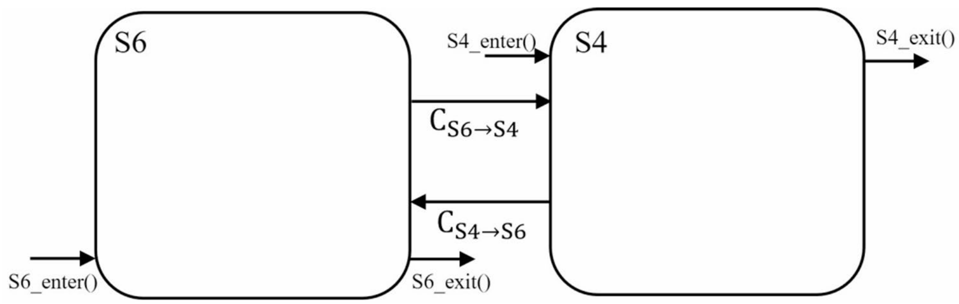

Fig. 6. The conversion relationship between the remote driving state and the vehicle driving exception handling state.

the key parameters of the SVM algorithm used in this paper are set to: the penalty parameter $c_s = 10$ and the kernel parameter $g_s = 0.01$.

Figure 5 displays the training result of the shift timing auxiliary decision model using the SVM algorithm. Figure 5(a) depicts the training result obtained on the training set. The training set contains a total of 1,836 shift process segments, including 954 reasonable decision segments of shift timing and 882 unreasonable decision segments of shift timing. The model is trained using 954 reasonable decision fragments, and the trained model has been shown to be able to make correct judgments on the rationality of 926 fragments, achieving an accuracy rate of 97.1%. Similarly, the model is trained using 882 unreasonable decision fragments, and the results have indicated that the trained model can make correct judgments on the rationality of 853 fragments, with a success rate of 96.7%. Figure 5(b) displays the prediction result obtained by the trained model on the test set. The test set contains a total of 325 shifting process segments, of which 166 are reasonable decision segments for shifting timing. Based on the results, the model can correctly predict 160 of the 166 reasonable decision fragments, having a correct rate of 96.4% and the prediction of unreasonable decision fragments of 97.5%. Thus, the auxiliary decision-making model of remote driving shift timing based on the SVM algorithm has high decision-making accuracy. Next, the model is applied to the shift control of tracked vehicle remote driving, which can effectively improve the shift smoothness of a vehicle.

Integrated shift control strategy for tracked vehicle power and transmission systems based on HFSM

To ensure that the vehicle controller can control each shift actuator to complete the shift process successfully and improve the smoothness of the vehicle shift, this study uses a hierarchical finite state machine to design the vehicle shift control system. The upper control system state mainly divides the driving control state of a tracked vehicle into the remote driving power transmission integrated shift control state, denoted by $S4$, and the shift exception handling state, denoted by $S6$. In state $S4$, a driver sends the shift instruction through the remote-control terminal. The vehicle controller controls the throttle, clutch, and shift control device to automatically realize the power transmission integrated shift control.

Figure 6 shows the conversion relationship between the remote driving state and the vehicle driving exception handling state. When a state transition condition $C_{S4\to S6}$ is satisfied, the vehicle controller will convert from state $S4$ to state $S6$, which can be expressed as follows:

Parameter	Meaning	Function
$flag_s$	Upper state flag	$flag_s$ is used to mark the upper state; the $flag_s$ values corresponding to different states are different.
$flag_s^{\mathrm{sub}}$	Lower state flag	In the sub-state of state $S4$, the $flag_s^{\mathrm{sub}}$ values corresponding to different shift sub-states differ.
$flag_g^c$	Shift instruction flag	$flag_g^c$ is used to mark the shift instruction, and its value indicates whether the vehicle controller receives different shift instructions, where zero represents that the empty gear instruction is received, one indicates the receipt of the starting instruction, two denotes the receiving of an upshift command, there stands for receiving the downshift instruction, six represents receiving the reverse gear instruction, and 11 represents keeping the original gear.
$flag_g^{\mathrm{err}}$	Flag of shift abnormality	$flag_g^{\mathrm{err}}$ is used to mark the cause of abnormal shift; different abnormal shift marks correspond to different abnormal shift treatment measures.
$flag_g^{\mathrm{suc}}$	Successful flag of gearing	$flag_g^{\mathrm{suc}}$ is used to mark whether the gearing is successful, and its value indicates different gearing states, where 1 represents the first shift failure back to the original gear, and 2 means that it is not possible to return to the original gear, take out the empty gear, and stop.

Table 2. The meanings and functions of the finite state machine control parameters.

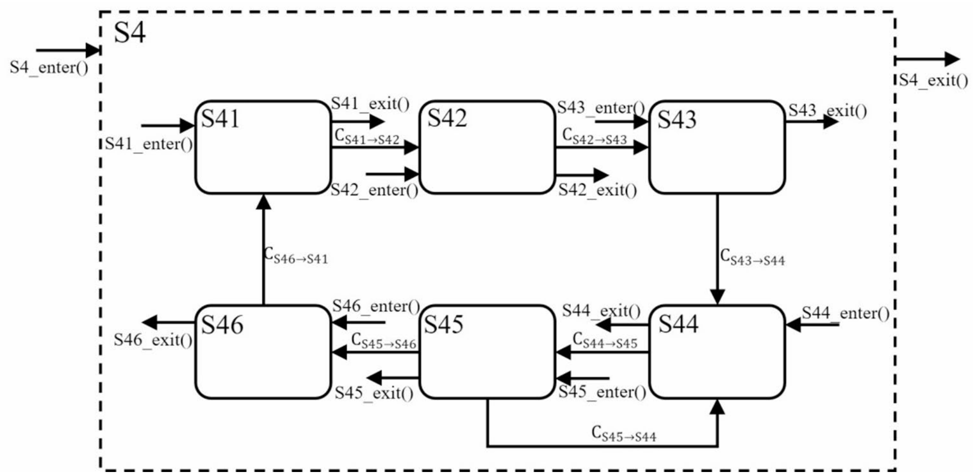

Fig. 7. The schematic diagram of the shift sub-state transition in state S4.

$$F(\mathrm{S6}, C_{\mathrm{S4}\rightarrow\,\mathrm{S6}}) = \mathrm{S4} \tag{14}$$

When the state generates the entry and exit actions, their functions are triggered accordingly. Namely, a specific function is triggered only once when the state produces the corresponding action, that is, the entry or exit action, which is used to assign the control variables and start and close the timer. Some control parameters for finite state machine control are shown in Table 2.

Design of shift control state machine based on shift timing auxiliary decision

Referring to a driver 's shift operation process, the remote driving integrated shift control state $S4$ is divided into six sub-states: the on-gear control state ($S41$), the shift timing auxiliary decision state ($S42$), the clutch separation state ($S43$), the off-gear control state ($S44$), the gear shift state ($S45$), and the clutch engagement state ($S46$).

Figure 7 displays the state transition diagram of the shift control sub-state in state $S4$, and the state transition conditions are defined as follows:

$$\begin{cases} F\left(\mathrm{S41,C_{S41\rightarrow\,S42}}\right) = \mathrm{S42} \\ F\left(\mathrm{S42,C_{S42\rightarrow\,S43}}\right) = \mathrm{S43} \\ F\left(\mathrm{S43,C_{S43\rightarrow\,S44}}\right) = \mathrm{S44} \\ F\left(\mathrm{S44,C_{S44\rightarrow\,S45}}\right) = \mathrm{S45} \\ F\left(\mathrm{S45,C_{S45\rightarrow\,S44}}\right) = \mathrm{S44} \\ F\left(\mathrm{S45,C_{S45\rightarrow\,S46}}\right) = \mathrm{S46} \\ F\left(\mathrm{S46,C_{S46\rightarrow\,S41}}\right) = \mathrm{S41} \end{cases} \tag{15}$$

In the S41 state, a tracked vehicle has successfully hung into the command gear. The driver has controlled the vehicle to achieve acceleration, deceleration, and steering in the remote control. When the driver needs to shift, he presses the remote control corresponding to the shift key to send the shift instruction. The vehicle controller monitors and collects the driver's control information in real time. When the driver's shift signal is judged to meet a state transition requirement $C_{\mathrm{S41}\rightarrow\,\mathrm{S42}}$, the control state enters state $S42$ from state $S41$. A variable $gear_{\mathrm{org}}$ in state $S41$ exits action function $\mathrm{S41\_exit}\,()$, which is responsible for recording the original gear, and $gear_{\text{org}} = gear_{\text{cur}}$. At the same time, the expected gear $gear_{exp}$ is obtained using the shift flag $flag_g^c$ as follows $gear\text{cmd}_{exp}$.

The $S42$ state is an auxiliary decision state of shift timing. After the vehicle controller receives the shift command, the shift timing decision is made, and the engine speed adjustment is performed. For the control of engine speed, the speed servo control based on the fuzzy PID algorithm[21] is adopted. In Sect. Data mining of shift timing decision based on K-Means algorithm, the manned driving data of tracked vehicles are mined, and it has been concluded that a driver generally increases the throttle opening and vehicle speed before performing the upshift operation. However, before the downshift operation, the operation of reducing the throttle opening and vehicle speed is generally conducted. An engine speed cluster center n_e^c, presented in Sect.Data mining of shift timing decision based on K-Means algorithm, is set as a target engine speed n_e^{exp}, that is, $n_e^{\text{exp}} = n_e^c$. In the process of engine speed adjustment, the shift timing auxiliary decision model monitors the values of engine speed n_e and throttle opening β_t in real time. When the model determines that the shift timing is reasonable, the vehicle enters the $S43$ state from the $S42$ state and starts the clutch separation operation. The specific operational process is illustrated in Fig. 8.

State $S43$ represents the clutch separation state. After entering this state, the vehicle controller controls the clutch hydraulic cylinder to perform the clutch separation operation. When a clutch displacement x_{clh} is larger than the preset value, the state transition condition $C_{\text{S43}\to\text{S44}}$ is satisfied, and the control state changes from state $S43$ to state $S44$.

After entering the $S44$ state, the vehicle controller controls the corresponding gear hydraulic cylinder to perform the operation of hanging the empty gear. When the vehicle is removed into the empty gear, the control state enters the $S45$ state from the $S44$ state. After entering the $S45$ state, the vehicle controller controls the shift actuator to change into the corresponding gear. At this time, the control state changes from state $S45$ to state $S46$. In addition, to solve the problem of too-long gearing caused by the top teeth of the synchronizer during the gearing process, this study sets the time variables t_s and t_g^{th} in $\text{S45\_enter}\ ()$, where t_s is used to calculate the gearing duration, and t_g^{th} is the gearing time threshold. When $t_s \geq t_g^{\text{th}}$, the system determines the abnormal shift operation. The control system returns from state $S45$ to state $S44$, and let $gear\text{org}_{exp}$. The tracked vehicle is removed into the neutral gear and returned to the original gear.

After entering the $S46$ state, the vehicle controller controls the clutch hydraulic cylinder to perform the clutch engagement operation, and after the clutch is combined, the controller enters the $S41$ state. The driver continues to control the tracked vehicle in the $S41$ state. The state transition rules of the power transmission integrated shift controller are presented in Table 3.

Design of shift fault handling state machine based on fixed time threshold method

To increase the driving safety of tracked vehicles, the state of each actuator during the shifting process is monitored in real time. By setting the time threshold t_s^{th} value, the smooth execution of each shift state is judged. The value of a time variable t_s is set in the action function of each state, and the t_s value indicates the

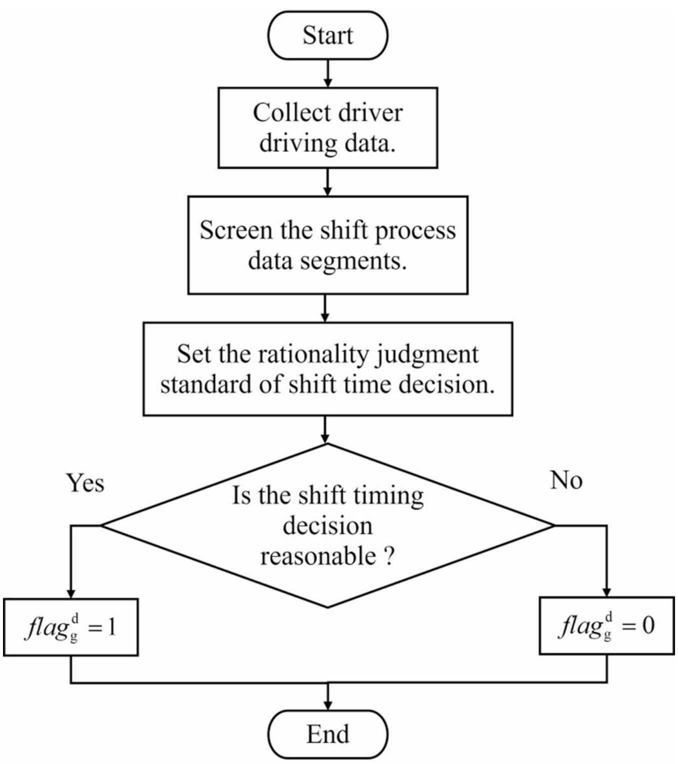

Fig. 8. The workflow of the shift timing auxiliary decision state.

State	$flag_s^{\mathrm{sub}}$	Transition condition	Condition setting	Target state
$S41$	41	$C_{\mathrm{S41}\rightarrow\ \mathrm{S42}}$	$flag_g^c \neq\ 11$	S42
S42	42	$C_{\mathrm{S42}\rightarrow\ \mathrm{S43}}$	$flag_g^d = 1$	S43
S43	43	$C_{\mathrm{S43}\rightarrow\ \mathrm{S44}}$	$x_{\mathrm{clh}} \geq\ 48\mathrm{mm}$	S44
S44	44	$C_{\mathrm{S44}\rightarrow\ \mathrm{S45}}$	$gear_{\mathrm{cur}} = 0$	S45
S45	45	$C_{\mathrm{S45}\rightarrow\ \mathrm{S44}}$	$t_s \geq\ 4\mathrm{s}, gear_{\mathrm{cur}} \neq\ gear_{exp}$	S44
		$C_{\mathrm{S45}\rightarrow\ \mathrm{S46}}$	$t_s < 4s, gear_{\mathrm{cur}} = gear_{exp}$	S46
S46	46	$C_{\mathrm{S46}\rightarrow\ \mathrm{S41}}$	$x_{\mathrm{clh}} \leq\ 1.8\mathrm{mm}$	S41

Table 3. The state transition rules from $S41$ to $S46$.

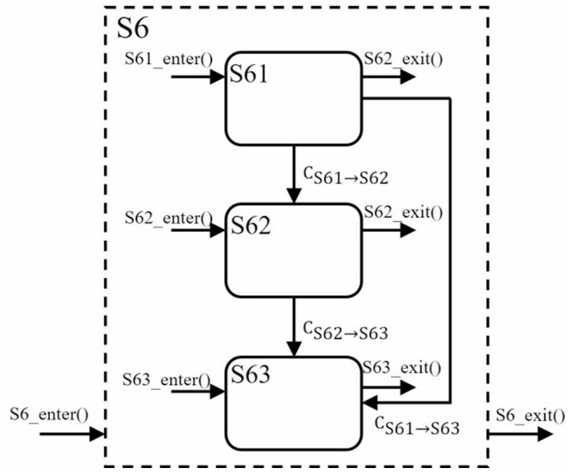

Fig. 9. The shifting exception handling state transition diagram.

State	$t_s^{\mathrm{th}}\ (s)$	$flag_s^{\mathrm{sub}}$	Specific treatment measures
$S43$	2	43	Brake and shut down the engine
$S44$	2	44	Separate the clutch
$S45$	3.2	45	Shift into the neutral gear
$S46$	2.5	46	Disengage the clutch and take off neutral

Table 4. The shift exception handling state transition rules.

execution time of a shift sub-state. When $t_s \geq\ t_s^{\mathrm{th}}$, the state execution time exceeds the preset state execution time threshold, and it can be considered that the actuator does not perform the shift action smoothly, which results in an abnormal shift. At this time, to ensure the ride comfort and safety of a vehicle, it is necessary to handle the abnormal shift. In the $S45$ state, to avoid frequent parking of a tracked vehicle, the controller returns the original gear after the first shift failure. If the time to change back to the original gear also exceeds the preset threshold, the shift of the vehicle is judged as abnormal, and the abnormal shift situation is handled accordingly.

Figure 9 depicts the schematic diagram of the shift exception handling sub-state transition process. As shown in Fig. 9, the shift exception handling state is divided into three sub-states: a clutch separation state ($S61$), a neutral gear state ($S62$), and a braking stop state ($S63$). The purpose of states $S61$ and $S62$ is to cut off the power rapidly when a tracked vehicle experience an abnormal shift. State $S63$ represents a vehicle braking parking state, and in this state, the vehicle controller takes different parking measures according to the particular causes of the abnormal shift. When the clutch separation fault occurs, aiming to address the difficulty of cutting off the engine's output power, emergency braking is performed, and the engine is extinguished. Considering the other reasons that can yield the cutting of the output power of the engine to avoid frequent start-up of the engine, this study does not perform the engine flameout treatment in the $S63$ state. Whether the brake stops or not is controlled by the driver sending the brake command at the remote-control end. The time threshold of each shift sub-state and the corresponding abnormal flags and shift exception handling measures are displayed in Table 4.

Experimental verification

The effectiveness of the proposed shift control strategy for tracked vehicles was verified by bench test and real vehicle test. The structure of the test bench is shown in Fig. 10. The test bench mainly includes hydraulic pump-motor mechanism, clutch, gearbox, brake and pneumatic load mechanism. The hydraulic pump-motor mechanism replaces the engine to generate power. The pneumatic load mechanism provides load for the transmission system and simulates the vehicle running on the ground. The hydraulic pump station provides power for each hydraulic actuator. Each transmission mechanism is equipped with a speed sensor to measure the speed of each component. In the tests, the vehicle controller received the remote control end shift command through an image transmission station and controlled the throttle, clutch, and other shift actuators for shift operation. After the bench test, we conducted a real vehicle test. Throughout the entire test process, the values of the vehicle's impact degree j_g were measured via sensors. The j_g were then compared with the impact degree standard for heavy tracked vehicles $\widehat{j}_g = 31.6\mathrm{m/s^3}$ to analyze the shift smoothness. Due to the inconvenient disclosure of some shape parameters of the vehicle, there is no picture of the real vehicle test scene in the paper. The way to obtain relevant data was explained at the end of the article. The control program of the vehicle controller was written in C++based on QT software, and the core processor is STM32F103 controller. The driving data recorder was responsible for recording vehicle driving data, which was convenient for vehicle data analysis and control optimization.

Bench test

Tracked vehicle neutral-to-second-gear shift control under no-load condition

The in-situ no-load shifting experiment of the tracked vehicle was performed. The process of the driver's remote control driving the tracked vehicle to complete the two-gear starting in the case of in-situ no-load is illustrated in Fig. 11. At a time $t_1 = 1.5s$, the driver sent a start command to the vehicle controller through the remote control terminal. At that time, $flag_g^c = 1$ satisfied a state transition condition $C_{\mathrm{S41} \to \mathrm{S42}}$. The shift control changed from state $S41$ to state $S42$, and $flag_s^{\mathrm{sub}} = 42$. In the $S42$ state, the vehicle controller realized the engine speed servo control, and the engine speed was significantly increased. The shift timing auxiliary decision model monitored the values of engine speed n_e and throttle opening β_t in real time. When $t_2 = 5.35s$ the auxiliary decision-making model of shift timing judged that the shift operation could be carried out at the current moment, changing from state $S42$ to state $S43$. At that moment, the vehicle controller was in the state control clutch separation. After the clutch was separated at $t_3 = 5.9s$, the vehicle controller controlled the vehicle to start gearing in the $S44$ state; $gear_{\mathrm{cur}} = -1$ indicated that the vehicle was in the state of hanging gear. At $t_4 = 8.2s$, the vehicle completed the block, and $gear_{cur} = 2$. After the vehicle controller controlled the clutch to combine at $t_5 = 8.87s$, the vehicle entered state $S41$ from state $S46$. After the vehicle started, the driver pushed the remote control rocker forward and issued refueling instructions to increase the speed.

Tracked vehicle 2nd-to-3rd gear shift control under no-load condition

Figure 12 showed the changes of parameters in the process of 2nd-to-3rd gear shift control of tracked vehicle under no-load condition. At $t_1 = 5.1s$, the driver sent an upshift command to the vehicle controller through the remote control terminal, and $flag_g^c = 2$. The shift control changed from state $S41$ to state $S42$. At that time, the shift timing auxiliary decision model judged that the engine speed $n_e$and the throttle opening β_t at the current moment met the shift requirements without the need for adjusting the engine speed and directly entered

Fig. 10. The test bench structure diagram.

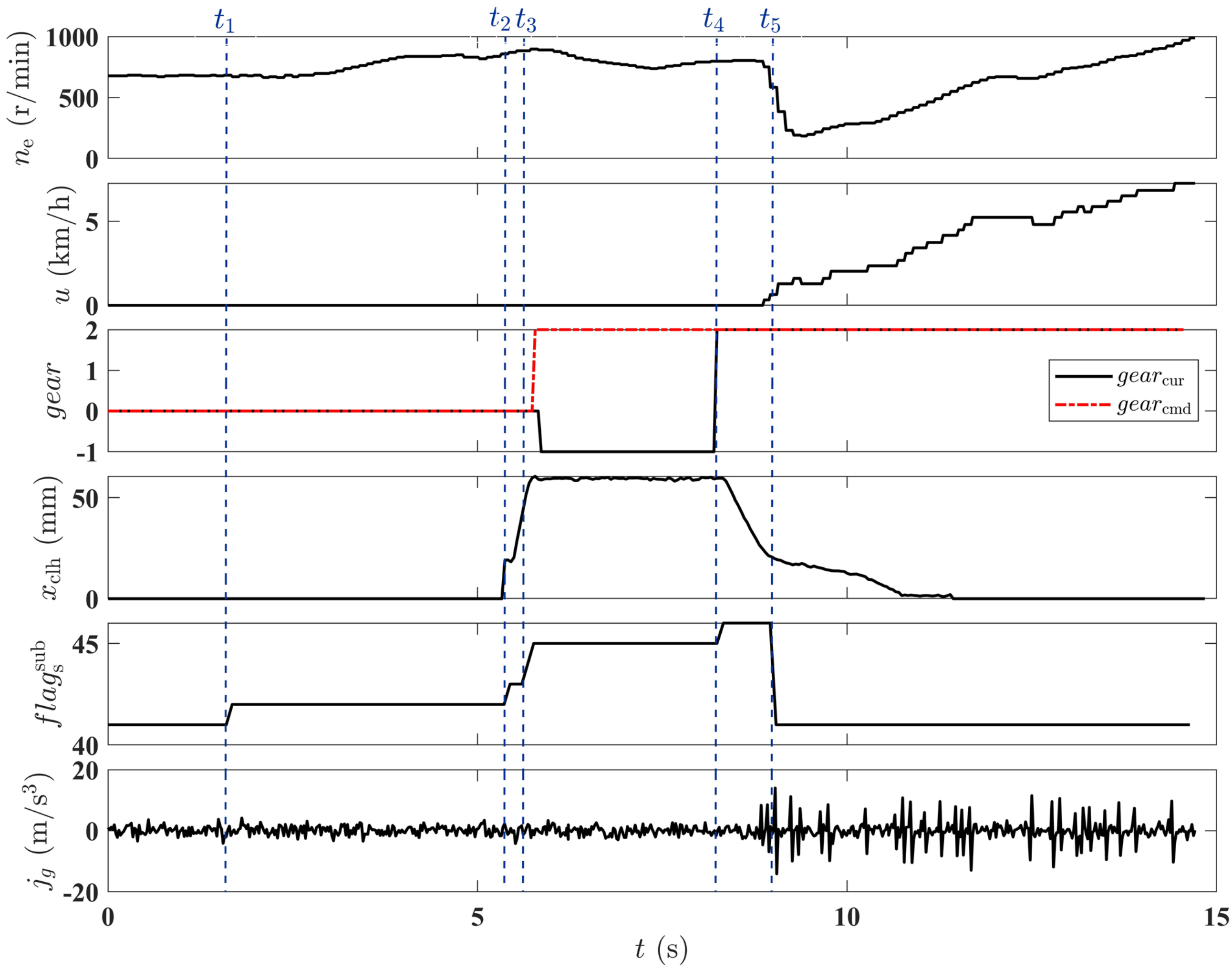

Fig. 11. The change of each parameter in the process of the second gear starting shift control of the tracked vehicle under no-load condition.

state $S43$ from state $S42$. At $t_2 = 8.3s$, the vehicle controller controlled the tracked vehicle to complete the upshift operation, and the whole shift process took 3.2 s. In the $S41$ state, the vehicle controller re-accepted the throttle and steering instructions of the remote control end to control the vehicle.

Tracked vehicle control process of returning to 4th gear after 4th-to-5th gear shift failure

The changing trend of each parameter when the vehicle controller controlled the tracked vehicle to perform the four- to five-gear operation is depicted in Fig. 13. The results indicated, that at $t_1 = 3.03s$, the driver sent an upshift command to the vehicle controller through the remote control terminal. The vehicle controller entered the $S42$ state and performed engine speed servo control to increase the engine speed to a suitable shift range. At $t_2 = 10.95s$, the shift timing auxiliary decision model judged that the current values of engine speed n_e and throttle opening $\beta_{\ t}$ state could meet the shift requirements, and $flag_g^d = 1$. The shift state changed from state $S42$ to state $S43$. The vehicle controller controlled the clutch separation and starts shifting. Further, at time $t_3 = 12.3s$, the vehicle controller entered the $S45$ state and began to hang five gears. In the synchronization process, the synchronizer failed, which increased the hanging time. At $t_4 = 15.3s$, the gearing time exceeded the preset time threshold, and the vehicle controller judged that the gearing failed. The vehicle controller controlled the vehicle to re-enter the $S44$ state and the hanging cylinder to return to the neutral gear and set $gear_{\text{cmd}} = gear_{\text{org}}$ in the exit action function S44_exit ()of the $S44$ state. After the vehicle controller controlled the tracked vehicle to return to the neutral gear, it re-entered the $S45$ state from the $S44$ state, and $gear_{\text{cmd}} = 4$. The vehicle controller controlled the vehicle to re-hang into four gears at $t_5 = 16.87s$. After the 4th gear was completed, the vehicle controller controlled the clutch to combine and return to the $S41$ state.

Tracked vehicle shift abnormal handling

In order to ensure the safety of the vehicle's remote driving, the experimental verification of the shift failure processing state machine developed in this paper was carried out. When the tracked vehicle performed shift control, the vehicle controller monitored the status of each shift actuator in real time. When an actuator fault was

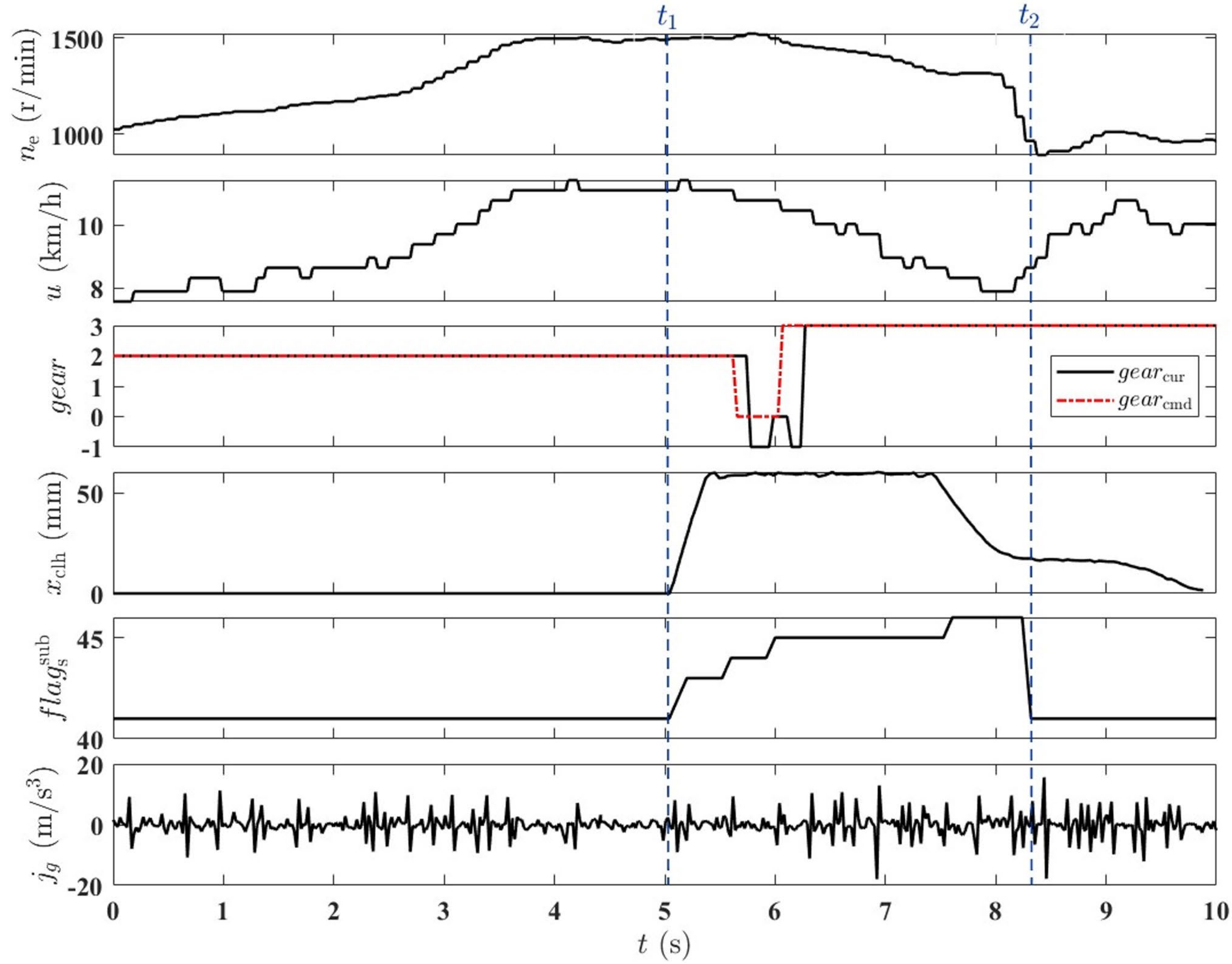

Fig. 12. The changes of parameters in the process of two-gear to three-gear control of tracked vehicles under no-load conditions.

judged to cause an abnormal shift, the vehicle controller conducted the corresponding measures timely to ensure driving safety. Considering an abnormal shift caused by clutch engagement failure, the developed abnormal shift handling strategy for tracked vehicles was verified. The control process of the vehicle to return to the neutral gear after the clutch combination failure was monitored by the vehicle controller during the starting process of the tracked vehicle, as shown in Fig. 14. When $t_1 = 2.9s$, the driver sent the vehicle start signal to the remote-control terminal, and the vehicle controller started to adjust the engine speed, made an auxiliary decision of the shift timing, and realized the clutch separation control. Due to the failure of the clutch cylinder displacement sensor, the sensor feedback signal showed that the clutch was in a separate state. The vehicle controller directly entered the state and controlled the hanging cylinder to carry out the two-gear operation. At $t_2 = 8.8s$, the vehicle was successfully hung into 2nd gear, and the vehicle controller started to control the clutch engagement. At $t_3 = 11.3s$, the vehicle controller judged the clutch engagement fault according to the feedback information obtained by the clutch cylinder displacement sensor. Next, the state of the upper control system was converted from the shift control state $S4$ to the shift exception handling state $S6$, $flag_s = 60$, $flag_s^{\mathrm{sub}} = 61$. At that time, the vehicle controller started to control the tracked vehicle to remove the empty gear parking operation and returned to the neutral gear at $t_4 = 11.9s$.

Real vehicle test

In this experiment, the tracked vehicle performed a remote driving continuous shift test. The continuous lifting gear operation speed, gear, and clutch displacement of the tracked vehicle are presented in Fig. 15. After the vehicle controller received the remote-control terminal's upshift command, it briefly increased the throttle opening, aiming to improve the speed of tracked vehicles, avoid excessive loss of vehicle speed due to long shift time, and improve the shift success rate. After receiving the downshift command, the vehicle controller employed the speed reduction method by reducing the throttle. The purpose was to avoid too-large differences between the spindle speed and the speed of the passive gear, as well as to decrease the synchronization time

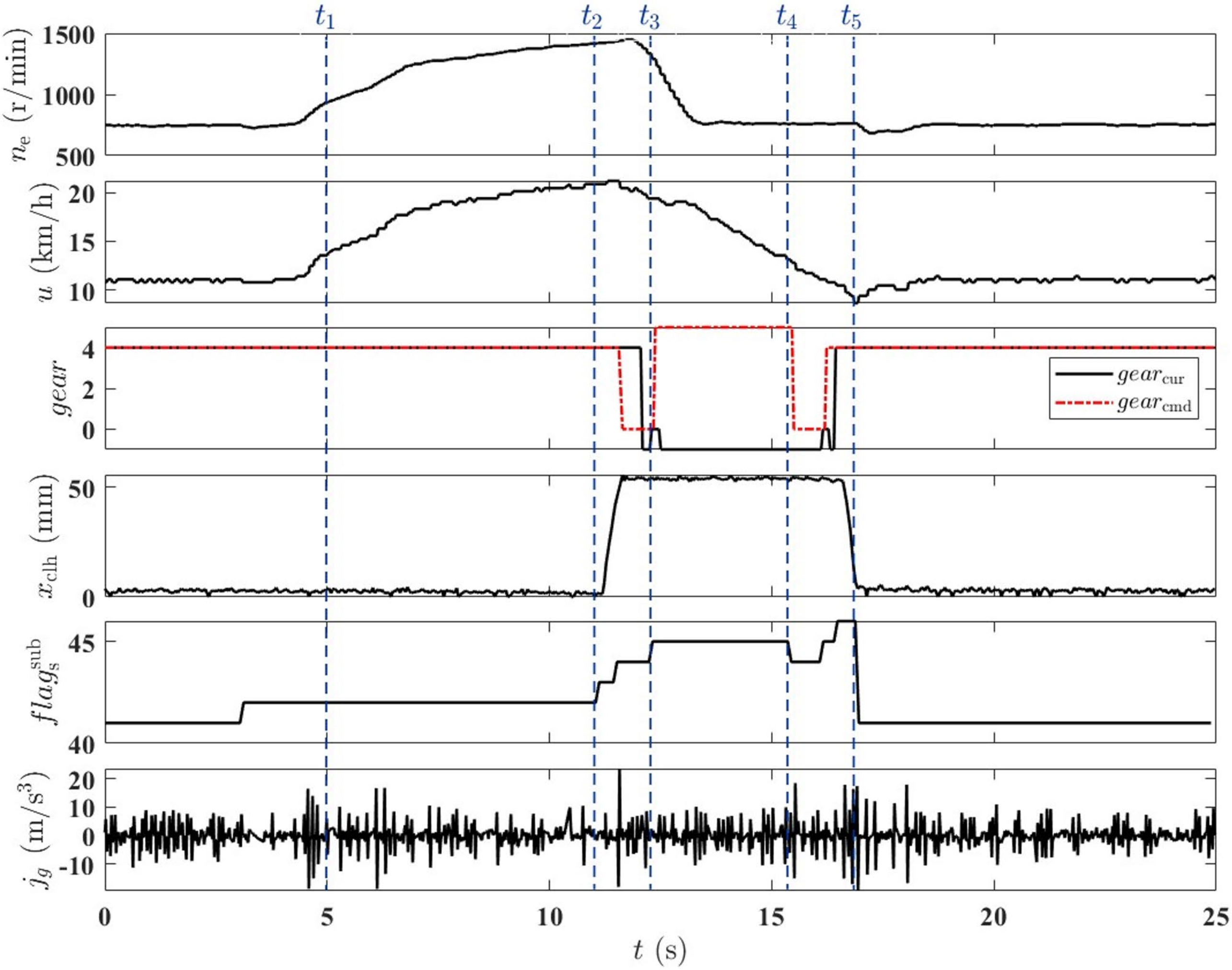

Fig. 13. The tracked vehicle's results in the situ test of no-load hanging five-gear failure and returned to the four-gear.

of the gear. The maximum speed of the tracked vehicle was 6.43 $\mathrm{m/s}$. The changing trend of the longitudinal impact degree of the tracked vehicle during the shift process is presented in Fig. 16. During the start-up and downshift of the tracked vehicle, the longitudinal impact of the vehicle was large, and the maximum impact was 19.23 $\mathrm{m/s^3}$ when the three-gear downshift was two-gear. The average longitudinal impact of the shift process of the tracked vehicle was 9.02 $\mathrm{m/s^3}$. The average shift control time of the entire shift process of the tracked vehicle was 4.6 s, which was close to the average shift control time of the manned vehicle, presented in Sect. 1.1. The test results showed that the proposed integrated shift control strategy of power transmission could coordinately control the shift actuator, achieving a smooth shift of the tracked vehicle.

Conclusion

In this paper, a remote driving shift control strategy for tracked vehicles based on driving data mining was proposed for unmanned heavy tracked vehicles, which effectively improved the shift smoothness of vehicles under remote driving conditions. Based on the test results, the following conclusions can be drawn:

(1) In this study, the manned driving data are deeply mined using the K-Means algorithm, and the SVM algorithm is employed to design an auxiliary decision-making model for the shift timing of tracked vehicle remote driving. The proposed model can effectively address the problem of low shift smoothness caused by the unreasonable shift timing decision when a tracked vehicle is driving remotely.

(2) The integrated shift control strategy of power transmission based on the finite state machine design can control each shift actuator to cooperate to achieve a shift. In the case of an unsuccessful shift caused by a synchronizer failure, the vehicle controller can control the shift actuator to return to the original gear automatically, thus avoiding a large speed loss caused by the vehicle. For the abnormal shift caused by the failure of a clutch combination, the vehicle controller controls the shift cylinder to return to the neutral gear timely, thus effectively improving the driving safety of the vehicle.

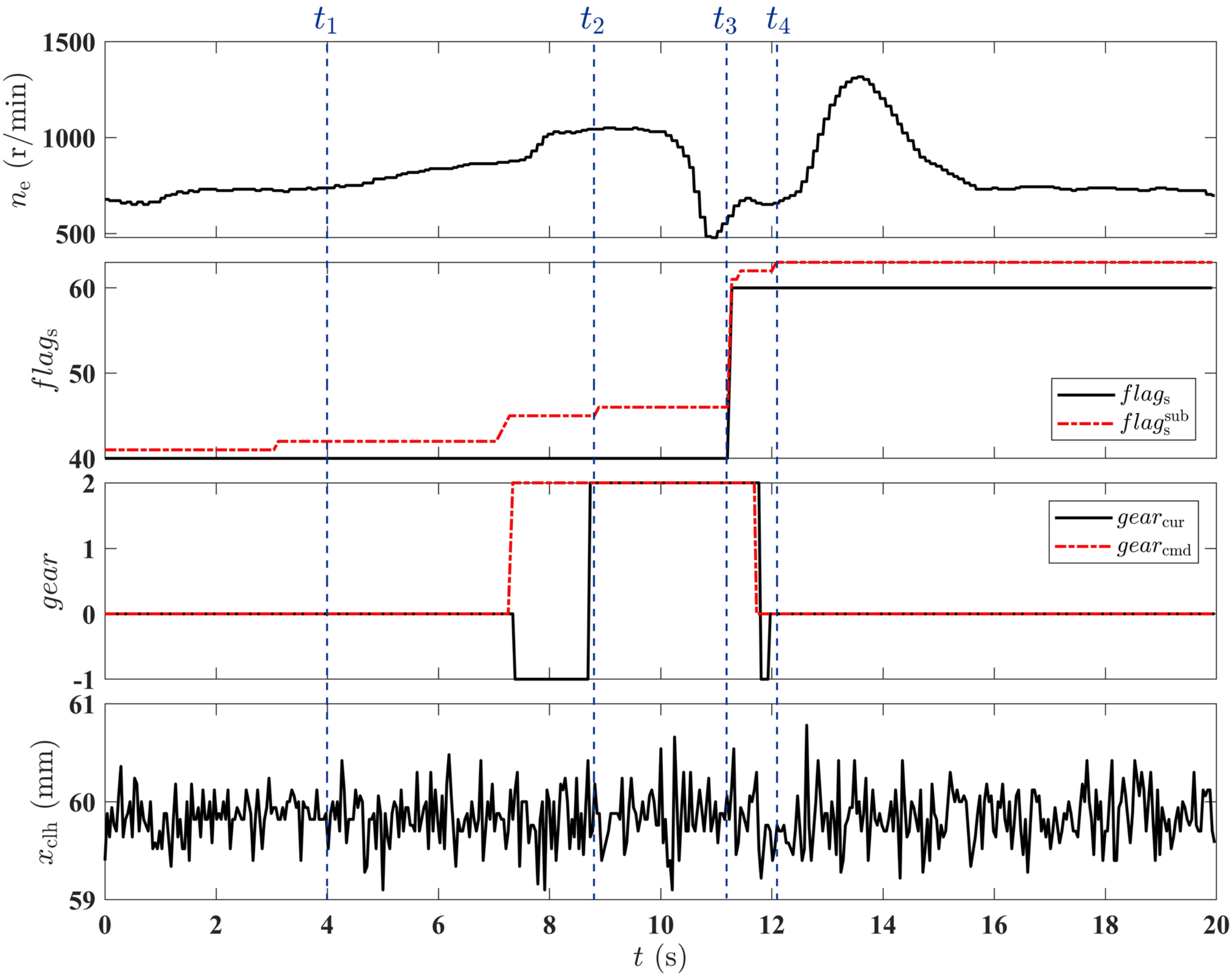

Fig. 14. The results in the situation when the clutch failure leads to an abnormal shift; the action of each actuator in the shift process is analyzed.

(3) Tracked vehicles shift continuously, and during the start-up and downshift processes of a tracked vehicle, the longitudinal impact of the vehicle is large, and the maximum impact is 19.23 $\mathrm{m/s^3}$ when the three and two gears are down. During the entire shifting process, the average shift impact of the tracked vehicle is 9.02 $\mathrm{m/s^3}$, the average shift time is 4.6 s, and the vehicle has good ride comfort.

(4) Due to constraints on research costs and conditions, we only used relatively mature control algorithms to control the vehicle. There is insufficient research on the application of advanced algorithms to the vehicle's drive system. The content of this study is a preliminary exploration of the driving control of heavy tracked vehicles undergoing unmanned modification. Integrating advanced algorithms such as deep learning into the existing control framework to improve the vehicle's control performance will be the focus of the next phase of research.

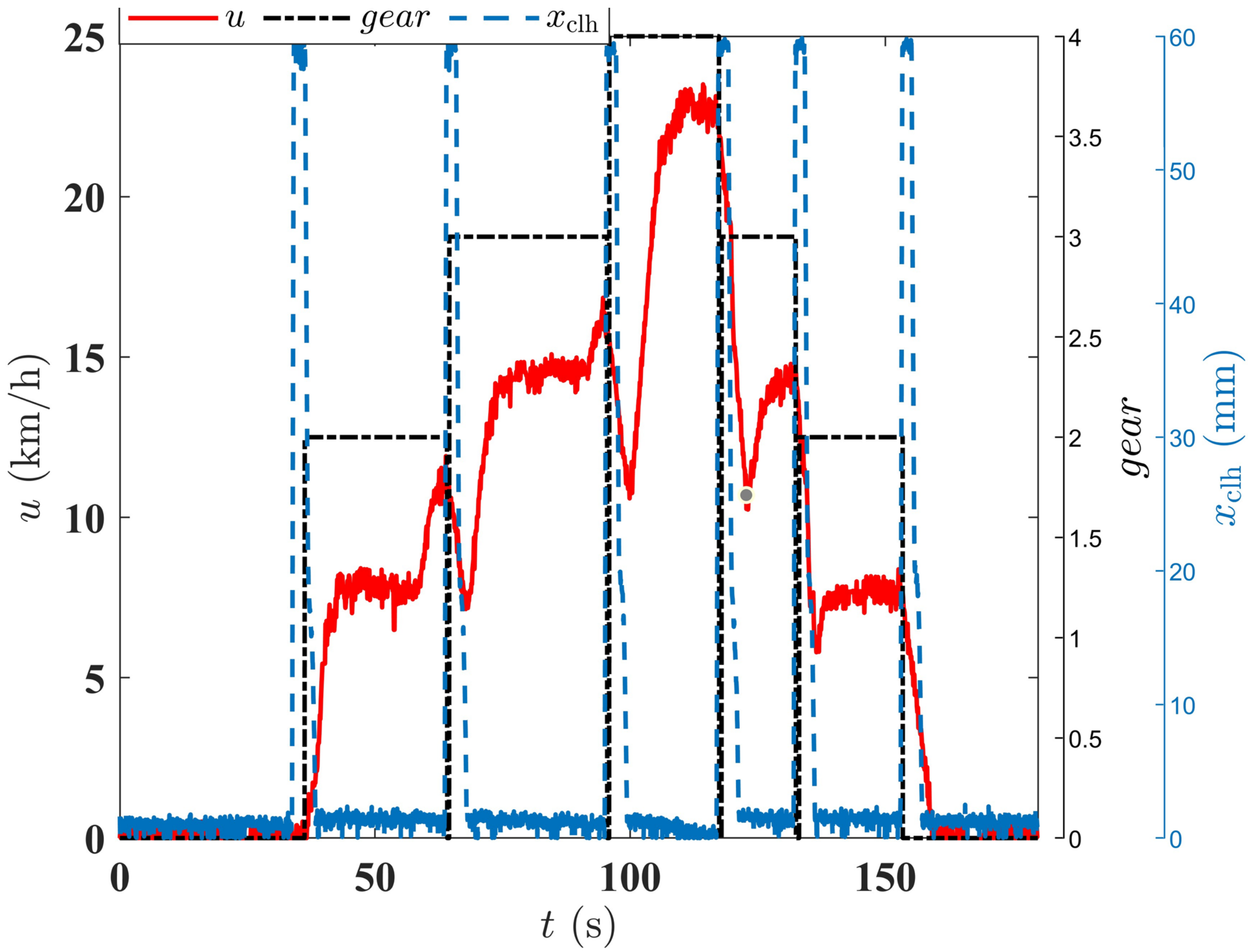

Fig. 15. The results obtained when the tracked vehicles continuously shifted the vehicle speed, gear, and clutch displacement changes.

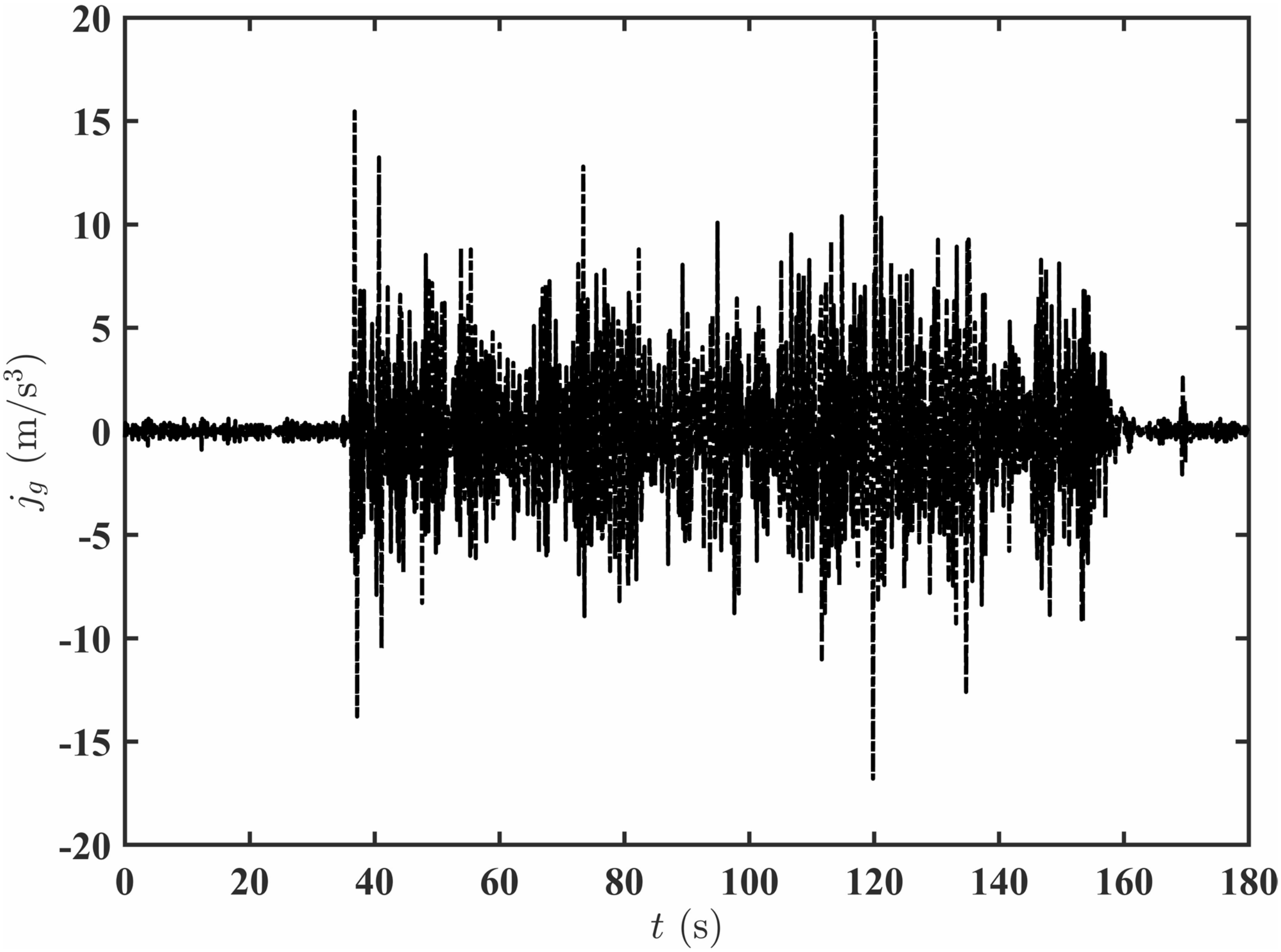

Fig. 16. The changing trend of the j_g value in the process of a continuous shift of a tracked vehicle.

Data availability
The datasets generated and analyzed during the current study are not publicly available due to secret vehicle parameters, but are available from the corresponding author (Weijian Jia) on reasonable request.

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Conceptualization, W.J. and H.Z.; Methodology, Y.Z. and W.J.; Software, W.J. and P.H.; Validation, Y.Z. and P.H.; Formal analysis, H.Z.; Investigation, P.H.; Data curation, H.Z.; Writing—original draft, Y.G. and W.J.; Writing—review & editing, W.J.; Project administration, Y.G. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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